

Complex AFNet: A Hybrid Complex-Valued Deep Network for Atrial Fibrillation Detection

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Abstract—Atrial fibrillation (AF) is the most common sustained cardiac arrhythmia, affecting nearly 59 million individuals worldwide by 2019 and significantly increasing stroke risk. This work presents a novel hybrid complex convolutional network for automated AF detection using Gramian angular field (GAF) transformations of electrocardiogram (ECG) signals. The proposed method converts 1-D ECG segments into 2-D image representations through Gramian angular summation field (GASF) and Gramian angular difference field (GADF) transformations, preserving temporal dependencies as geometric relationships in polar coordinates. Our proposed model treats GASF and GADF as real and imaginary components of complex-valued inputs, enabling phase-aware feature extraction through the proposed hybrid architecture combining real-valued and complex-valued convolutional layers. The network incorporates complex convolutions, modified rectified linear unit (ReLU) activations, and residual connections to capture subtle magnitude-phase interactions critical for detecting AF episodes. Evaluated on the PhysioNet AF dataset containing 1436 ECG recordings, this model achieves superior performance with 95.63% test accuracy, 96.43% precision, 92.05% recall, and 94.15% F1-score, outperforming state-of-the-art models by +1.52% accuracy and +2.1% F1-score. Interpretability analysis using Grad-CAM and integrated gradients (IGs) reveals class-specific attention patterns, with the model focusing on GADF for normal rhythm detection and GASF for AF classification, demonstrating clinically relevant pattern recognition capabilities of our model. To facilitate reproducibility and further research, the implementation of the proposed C-AFNet and the associated preprocessing and evaluation scripts are publicly available at this link.

Impact Statement—This work proposes a hybrid complex convolutional network for the automated detection of AF, a cardiac arrhythmia affecting over 59 million people globally and associated with a five-fold increased risk of stroke. By transforming raw ECG signals into GAF representations, the model leverages both magnitude and phase information through complex-valued

convolutions to capture subtle temporal dynamics. The approach achieves a high detection accuracy of 95.6%, outperforming state-of-the-art methods. Beyond accuracy, the model's interpretability is enhanced through Grad-CAM and IGs, offering clinically meaningful visualizations of decision rationale. This combination of performance and transparency makes the proposed method a promising tool for early, noninvasive cardiac screening in both clinical and remote healthcare settings, particularly where access to specialized diagnostic expertise is limited.

Index Terms—Atrial fibrillation (AF) detection, complex-valued neural networks (CVNN), deep learning, electrocardiogram (ECG) analysis, explainable artificial intelligence (AI), Gramian angular field (GAF).

I. INTRODUCTION

ATRIAL fibrillation (AF) is the most common sustained cardiac arrhythmia, with global cases rising from 33.5 million in 2010 to nearly 59 million by 2019 [1]. Stroke is a leading cause of mortality and disability worldwide, affecting over 1.6 million adults annually in India [2], causing 0.19 million deaths in the United States in 2021 [3] and 2.19 million deaths in China in 2019 [4]. AF increases the risk of ischemic stroke fivefold, making early detection crucial [5]. AF is categorized into several types based on duration and persistence: paroxysmal AF (self-terminating within seven days), persistent AF (lasting longer than seven days), and permanent AF (continuously present) [6]. The electrocardiogram (ECG) is the gold-standard diagnostic tool, where AF manifests as irregular R-R intervals and the absence of distinct P-waves, often replaced by low-amplitude fibrillatory (f) waves. Fig. 1 illustrates typical ECG patterns for normal sinus rhythm, persistent AF, and the particularly challenging paroxysmal AF.

Automated detection of AF from ECG signals remains a critical challenge, particularly for paroxysmal AF episodes that are intermittent, short-duration, and easily missed during routine clinical assessments. Despite significant advances in deep learning for ECG analysis, existing AF detection approaches face several fundamental limitations. Traditional methods that rely on handcrafted rhythm features [e.g., R-R interval (interval between two successive R-waves)] often fail to capture subtle morphological patterns indicative of AF. While recent deep learning models have demonstrated improved performance, most operate in the real-valued domain and process ECG signals either in their raw form or through frequency-domain transformations. These approaches typically treat complementary signal representations such as amplitude and phase information

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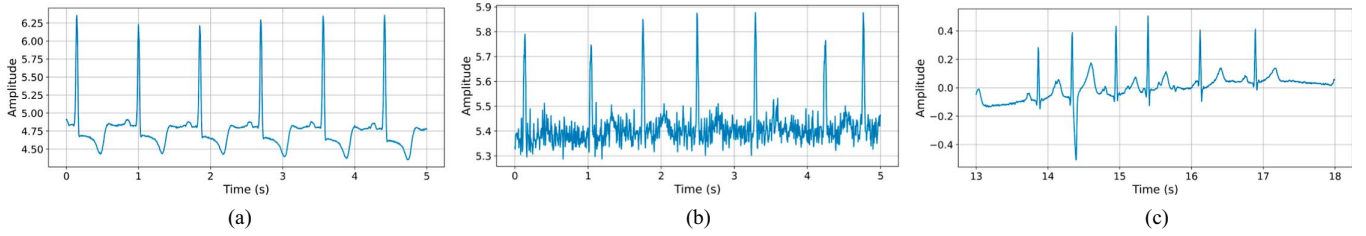


Fig. 1. Examples of a normal ECG and two types of AF ECGs: persistent and paroxysmal. (a) Normal. (b) Persistent AF. (c) Paroxysmal AF.

as separate channels or through simple concatenation, thereby overlooking the inherent coupling between these fundamental components. This methodological gap becomes particularly pronounced when detecting paroxysmal AF, where transient amplitude–phase relationships contain critical discriminative information. To address these limitations, we propose complex-AFNet (C-AFNet), a novel hybrid deep learning framework that explicitly models amplitude–phase coupling through complex-valued representations of ECG signals. Our approach transforms 1-D ECG segments into Gramian angular field (GAF) images, specifically Gramian angular summation field (GASF) and Gramian angular difference field (GADF), which are then treated as the real and imaginary components of complex-valued inputs. By employing a hybrid architecture that combines real-valued convolutional layers for efficient feature extraction with complex-valued layers for phase-sensitive modeling, our method captures interdependent temporal dynamics that conventional real-valued networks cannot fully exploit. The primary contributions of this work are as follows.

- 1) We introduce an innovative complex-valued deep learning framework for detecting AF, utilizing GASF and GADF representations of ECG signals as real and imaginary parts. This approach allows the model to effectively capture interactions between magnitude and phase.
- 2) We developed an ECG-to-image preprocessing pipeline that segments raw ECG signals into overlapping windows and encodes each into GAF-based representations. This approach enhances temporal continuity and maintains class balance, especially improving the detection of underrepresented paroxysmal AF segments.
- 3) We validate our method using a clinically annotated dataset, achieving superior performance over state-of-the-art methods, and propose a custom metric for evaluating training dynamics, revealing insights into model convergence and robustness.
- 4) To enhance clinical relevance, we employ post hoc interpretability techniques such as Grad-CAM and integrated gradients (IGs) in the image domain, highlighting the model’s attention to relevant regions. For easier reproducibility, codes and datasets will be made publicly available.

II. RELATED WORKS

Our work on the hybrid complex convolution network for AF detection integrates three primary domains: complex-valued

neural networks (CVNNs), GAF embeddings, and deep learning for AF detection. In the following, we critically review these areas to highlight the methodological gaps our architecture addresses. Additional citations are integrated throughout the methodology and experimentation sections.

A. CVNNs

CVNNs are effective for phase-sensitive tasks like radar, audio, and biomedical signal processing. Trabelsi et al. [7] established the mathematical foundation for deep complex networks; their focus remained on general image and audio classification. Bassey et al. [8] identified that the primary bottleneck for these networks in high-dimensional signal tasks is the lack of specialized normalization and residual links. We address these architectural gaps by introducing a staged complex-residual framework. Although Guberman [9] proposed activations such as zReLU to handle the lack of ordering in the complex field, such functions restrict activation to a single quadrant. In the context of ECG-based GAF images, where phase interactions are more erratic, we utilize ModReLU. This choice allows the network to suppress low-magnitude baseline noise critical to detect paroxysmal AF while retaining the full range of discriminative phase information. Expanding on the findings of Cole et al. [10], which demonstrated the superiority of CVNNs for phase-dependent tasks such as magnetic resonance imaging reconstruction, our study transitions the focus from reconstruction to diagnostic classification. By expressing GASF and GADF as real and imaginary components, we present the inaugural hybrid model adept at inherently capturing the amplitude-phase interactions crucial for identifying minor cardiac arrhythmias. Despite their potential, CVNNs have seen limited adoption in ECG analysis, particularly in modeling the interplay between amplitude and phase representations derived from time-series embeddings.

B. GAF for Time-Series Analysis

Time-series signals can be converted into images using GAF representations, which preserve temporal patterns via angular encoding. Wang and Oates [11] introduced GASF and GADF to capture cosine and sine-based dynamics. Prior work [12] used GASF and GADF as separate channels, ignoring their phase relationship. We instead treat them as the real and imaginary parts of complex inputs, enabling complex convolutions to model interdependent dynamics. Unlike recent real-valued

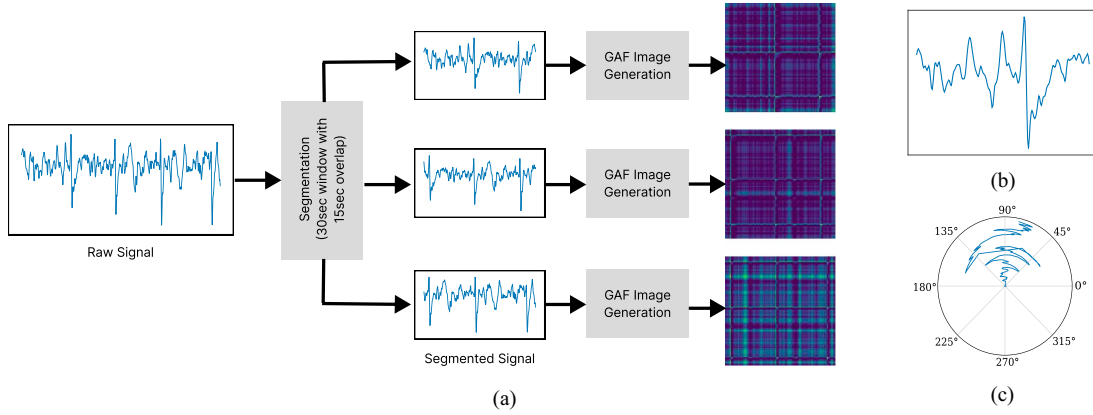


Fig. 2. (a) Overview of the process for transforming a 1-D ECG time series into 2-D GAF representations. (b) Normalization of the signal to the range $[-1, 1]$. (c) Angular encoding of the normalized signal in the polar coordinate space.

attention models [13], our approach retains critical phase information, marking the first use of CVNNs for GAF-based ECG AF detection. This formulation allows explicit modeling of amplitude–phase coupling, which is often overlooked in traditional image-based time-series classification.

C. Machine Learning and Deep Learning for AF Detection

Early AF detection relied on RR-interval and heart rate variability features with classical classifiers [support vector machine (SVM), random forest, and AdaBoost], often exceeding 90% accuracy on curated datasets (e.g., Kennedy et al. [14] and Rao et al. [15]). Deep learning soon dominated: Hannun et al. [16] achieved cardiologist-level accuracy with a 34-layer convolutional neural network (CNN) on PhysioNet [17], while RNNs [18], seq-to-seq models [19], and spectrogram CNNs [20] captured temporal and frequency–time features. Wavelet-based CNNs (e.g., Sun et al. [21]) demonstrated accuracy above 91% and an F1 score of approximately 89%. Attention and transformer–CNN hybrids further improved cross-dataset generalization, with Kraft et al. [22] showing strong results using a compact ConvNeXt-style 1-D CNN. Recent years have seen increasing adoption of transformer-based architectures for AF detection, leveraging self-attention to capture long-range dependencies in ECG signals. Li et al. [23] proposed SEResUTer, a U-Net-based model with ResNet modules and transformer encoders for ECG delineation and AF detection, achieving strong F1 performance across multiple databases. Hu et al. [24] combined residual blocks with a transformer encoder to detect paroxysmal AF and its onset/offset, demonstrating robust results on public datasets. Gündüz and Talu [25] evaluated spectral features, P waves, and R–R intervals using BiLSTM and convolutional long short-term memory models, reporting the best test accuracy with P-wave-based classification on the CPSC2021 dataset [26]. Recent studies also explore diverse input representations and architectures. Xie et al. [20] employed continuous wavelet transform (CWT) with a multi-branch ResNet to address class imbalance in single-lead ECGs, while Rahman et al. [27] fused raw ECG with discrete wavelet

transform (DWT) features in a lightweight 1-D CNN, highlighting the benefits of joint time-domain and time–frequency analysis. Aldughayfiq et al. [28] integrated ECG and photoplethysmogram (PPG) signals using a hybrid CNN–BiLSTM, demonstrating the effectiveness of multimodal fusion. In parallel, raw ECG-based approaches such as RawECGNet [29] and ArNet-ECG [30] emphasize improved generalization and robustness by directly learning morphological patterns across diverse recording conditions. Most existing AF detection methods rely on real-valued fusion strategies that do not explicitly model amplitude–phase coupling, motivating the use of unified complex representations in this work.

III. METHODOLOGY

A. Transforming ECG Signals into Complex-Valued Inputs

For effective AF diagnosis from quasi-periodic ECG signals [31], we employ GAF image transformations, which overcome the limitations of conventional methods. Unlike recurrent neural networks [32] that struggle with long-range signal dependencies, the GAF’s holistic image representation captures the full temporal context [33]. Furthermore, compared to time-frequency methods such as spectrograms and wavelets, GAF better preserves the morphological patterns critical for AF detection. By encoding temporal data as structured textures in a polar space, GAF enables CNNs to learn more robust features and achieve superior classification performance [34], [35], [36]. As illustrated in Fig. 2(a), generating GAF images from ECG signals involves several stages. The input raw ECG signal is first segmented into 30-second windows with a 15-second overlap, following the recommendations of the European Society of Cardiology Guidelines [6]. Next, the GAF transformation is applied to each segment, yielding both GASF and GADF representations [37]. Let the input time series be denoted as $X = \{x_1, x_2, \dots, x_N\}$. The transformation involves the following steps. First, the time series is *normalized* to the range $[-1, 1]$ to ensure consistent scaling across different recordings, as depicted in Fig. 2(b). The normalization is performed using

$$x'_i = 2 \times \frac{x_i - \min(X)}{\max(X) - \min(X)} - 1. \quad (1)$$

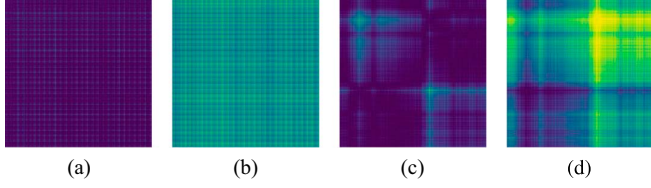


Fig. 3. GAF image representations of ECG segments. (a) GASF of a normal rhythm. (b) GADF of a normal rhythm. (c) GASF of an AF segment. (d) GADF of an AF segment.

Second, the *angular encoding* step maps the normalized signal values to the angular domain by computing the angle ϕ_i for each sample, as illustrated in Fig. 2(c)

$$\phi_i = \arccos(x'_i), \quad \phi_i \in [0, \pi]. \quad (2)$$

Third, the GAF matrices are computed to capture temporal dependencies. The *GASF matrix* is computed using the cosine of the sum of angles

$$\begin{aligned} \text{GASF}(i, j) &= \cos(\phi_i + \phi_j) \\ &= x'^T \cdot x' - \sqrt{1 - (x')^2}^T \cdot \sqrt{1 - (x')^2} \end{aligned} \quad (3)$$

as shown in Fig. 3(a) and (c). The *GADF matrix* is derived using the sine of the angle differences

$$\begin{aligned} \text{GADF}(i, j) &= \sin(\phi_i - \phi_j) \\ &= \sqrt{1 - (x')^2}^T \cdot x' - x'^T \cdot \sqrt{1 - (x')^2} \end{aligned} \quad (4)$$

as illustrated in Fig. 3(b) and (d). GAF matrices can be visualized as RGB images, as the matrices contain values in the range $[-1, 1]$, which are normalized to $[0, 255]$ and mapped to 8-bit pixel intensities. These normalized matrices are then stacked into RGB channels to form composite images suitable for CNN-based classification. As illustrated in Fig. 3, normal ECG segments exhibit regular, structured diagonal patterns, whereas AF segments display irregular, fragmented, and chaotic textures. Finally, to capture magnitude and phase dynamics simultaneously, the complex-valued matrix $\mathbb{Z} \in \mathbb{C}^{n \times n}$ is made by designating GASF as the real part and GADF as the imaginary part: $\mathbb{Z} = \text{GASF} + i \cdot \text{GADF}$.

B. Our Proposed C-AFNet

To utilize both amplitude and phase information in ECG representations, simultaneously & sequentially (in both ways), we proposed a hybrid architecture combining real-valued & complex-valued convolutional layers in an alternate manner. It is the first of its kind in its hybrid nature. This architecture incorporates residual connections [38] for deeper networks and efficient gradient flow, along with 2-D convolutional layers (Conv2D) for spatial feature extraction, batch normalization (BN) [39] to stabilize activations, and rectified linear unit (ReLU) activations [40] to introduce nonlinearity

$$\text{Output} = F(x) + x \quad (5)$$

where F represents a sequence of Conv-BN-ReLU operations. If the input and output dimensions differ, a 1×1 convolutional

projection is used to match them. To ensure stable gradient propagation during training, we use He et al.'s initialization strategy [41]. All complex layers are built from real operations on the real and imaginary parts, but their combined behavior aligns with true complex algebra.

1) *Complex Convolution*: Let the input and kernel be $z = x_r + ix_i$, $w = w_r + iw_i$, where $x_r, x_i, w_r, w_i \in \mathbb{R}$

$$w * z = (w_r * x_r - w_i * x_i) + i(w_r * x_i + w_i * x_r). \quad (6)$$

This complex convolution is implemented using four real convolutions. By preserving both real and imaginary cross terms, this layer captures phase interactions crucial for GASF and GADF image pairs [42], [43].

2) *Complex Batch Normalization*: We apply BN [39] separately to the real and imaginary channels

$$\hat{x}_r = \frac{x_r - \mu_r}{\sqrt{\sigma_r^2 + \epsilon}}, \quad \hat{x}_i = \frac{x_i - \mu_i}{\sqrt{\sigma_i^2 + \epsilon}}. \quad (7)$$

This reduces internal covariate shift and accelerates convergence, aligning with CVNN best practices [39].

3) *ModReLU Activation*: ModReLU [42], [44] operates on the magnitude of z while preserving its phase

$$\text{ModReLU}(z) = \text{ReLU}(|z| + b) \frac{z}{|z| + \epsilon} \quad (8)$$

here b is a learnable bias and the magnitude is given by $|z| = \sqrt{x_r^2 + x_i^2}$. Intuitively, small-magnitude activations are suppressed, while the original phase $\arg(z)$ is retained. The term ϵ is a small positive constant added to avoid numerical instability (division by zero) when $|z|$ is close to zero.

4) *Limitations of Standard Complex Derivatives*: Classical complex differentiation requires holomorphic functions, i.e., those satisfying the Cauchy–Riemann equations

$$\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y}, \quad \frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x} \quad (9)$$

for $f(z) = u(x, y) + iv(x, y)$. However, operations like ModReLU, separate BatchNorm, magnitude-based computations, and the use of real-valued loss function depend on both z and its conjugate $\bar{z}$, and these operations violate the Cauchy–Riemann conditions and are inherently nonholomorphic [42].

5) *Wirtinger Calculus for Complex Backpropagation*: To enable gradient-based optimization in nonholomorphic settings, Wirtinger calculus [45] treats the complex variable $z = x + iy$ and its conjugate $\bar{z} = x - iy$ as independent, allowing partial derivatives and supporting standard backpropagation

$$\frac{\partial f}{\partial z} = \frac{1}{2} \left(\frac{\partial f}{\partial x} - i \frac{\partial f}{\partial y} \right), \quad \frac{\partial f}{\partial \bar{z}} = \frac{1}{2} \left(\frac{\partial f}{\partial x} + i \frac{\partial f}{\partial y} \right) \quad (10)$$

where $f(z) = u(x, y) + iv(x, y)$. These definitions allow us to differentiate real-valued functions of complex variables, even when they are nonholomorphic. For a real-valued loss function $\mathcal{L}(z, \bar{z})$, the overall gradient with respect to z is as follows:

$$\nabla_z \mathcal{L} = \left(\frac{\partial \mathcal{L}}{\partial x} + i \frac{\partial \mathcal{L}}{\partial y} \right) = 2 \frac{\partial \mathcal{L}}{\partial \bar{z}}. \quad (11)$$

This formulation preserves the necessary gradient information for nonholomorphic mappings, enabling rigorous application of the chain rule during backpropagation.

6) *Gradient Flow in Practical Implementation:* In our implementation, each complex parameter $w = w_r + iw_i$ is represented by two real-valued tensors: $w_r, w_i \in \mathbb{R}^n$. Operations such as convolution, BN, and activation are applied separately to the real and imaginary components based on complex arithmetic. During training, the PyTorch autograd engine [46] computes the gradients of the real-valued loss $\mathcal{L}$ independently with respect to w_r and w_i : $(\partial\mathcal{L}/\partial w_r), (\partial\mathcal{L}/\partial w_i)$. This automatic differentiation mirrors the Wirtinger formulation, where the complex gradient is implicitly represented as

$$\nabla_w \mathcal{L} = \frac{\partial \mathcal{L}}{\partial w_r} + i \frac{\partial \mathcal{L}}{\partial w_i}. \quad (12)$$

As a result, performing manual calculations of Wirtinger derivatives is not needed.

7) *Final Output and Loss Computation:* Following a sequence of complex-valued operations, the final feature representation $z \in \mathbb{C}$ is transformed into a real-valued tensor by computing its magnitude: $|z| = \sqrt{\text{Re}(z)^2 + \text{Im}(z)^2}$. Using this magnitude, a fully connected layer generates a scalar logit $\hat{y} \in \mathbb{R}$ to calculate the binary classification loss via

$$\mathcal{L}(\hat{y}, y) = -y \log \left(\frac{1}{1 + e^{-\hat{y}}} \right) - (1 - y) \log \left(\frac{e^{-\hat{y}}}{1 + e^{-\hat{y}}} \right). \quad (13)$$

As all outputs are real-valued and operations are differentiable, the network is fully trainable using PyTorch's autograd [46]. The proposed hybrid architecture accepts paired GASF (real) and GADF (imaginary) inputs of size 224×224 , which are initially processed through real-valued convolutional blocks (A–E, Table I) to extract modality-specific features. These are fused into complex tensors and passed through complex residual blocks (F onward in Table I) to capture phase-aware interactions. Subsequent real-valued layers (blocks H–K) refine high-level features before final complex activations, enabling spatial and phase information integration. Unlike standard ResNet-18, our architecture omits the initial max-pooling layers [47] to preserve fine temporal localization, which is vital for detecting paroxysmal AF in GAF-encoded ECGs [48]. This hybrid design balances spatial precision with efficient phase-aware representation learning. Our complex architecture of the proposed hybrid C-AFNet is given in Fig. 4 and Table I.

C. Proposed Custom Metrics

To complement standard metrics, we propose a composite evaluation scheme capturing training dynamics and generalization. It comprises the following.

1) Stability Metric (SM):

$$\text{SM} = \mu - \lambda \cdot \sigma^2 \quad (14)$$

here, μ and σ^2 denote the mean and variance of validation accuracies per epoch considered together in an array, respectively, and λ controls the variance penalty. The metric, inspired by the bias–variance tradeoff [49], [50], favors consistent generalization.

2) Overfitting Penalty (OP):

$$\text{OP} = \exp(-\delta \cdot (\text{Acc}_{\max} - \text{Acc}_{\text{final}})) \quad (15)$$

here, $\text{Acc}_{\max}$ and $\text{Acc}_{\text{final}}$ denote the peak and final validation accuracies, respectively, and δ is a scaling factor. The metric penalizes overfitting by quantifying the generalization gap [51], [52].

3) Learning Speed (LS):

$$\text{LS} = \left(1 - \frac{e_{\text{peak}}}{e_{\text{total}}} \right) \cdot (0.09 \cdot \text{Acc}_{\text{peak}}) \quad (16)$$

here, e_{peak} is the peak epoch and e_{total} is the total epoch; and the metric reflects *convergence speed* [53], [54].

To consolidate these evaluations, we compute a weighted aggregate termed the forecast stability score (FS)

$$\text{FS} = \alpha \cdot \text{SM} + \beta \cdot \text{OP} + \gamma \cdot \text{LS} \quad (17)$$

where α , β and γ are the weighting coefficients that reflect the relative importance of stability, generalization, and speed, respectively. The FS enables a unified model comparison across training dynamics and performance.

IV. EXPERIMENTS AND RESULTS

A. Dataset Description

For robustness analysis, the proposed method is evaluated on three datasets, including out-of-distribution testing. The primary dataset used in this work is from the 4th China Physiological Signal Challenge 2021, titled *Paroxysmal AF Events Detection from Dynamic ECG Recordings* [26]. It comprises 1436 Lead I and II ECG recordings from 105 subjects, with each subject contributing 2–29 recordings, each lasting between 8.375 and 24 666.625 s. For recordings shorter than 30 s and for incomplete edge segments, zero-padding is applied at the end of the signal to obtain fixed-length segments. This ensures uniform input dimensionality across all ECG segments prior to GAF transformation. The dataset contains 732 normal, 475 persistent AF, and 229 paroxysmal AF samples, providing a balanced distribution. From this dataset, 115 010 GASF and GADF images were generated for each representation. The data were split into training (83 799), validation (20 158), and testing (11 053) sets. To evaluate generalization, trained models were further tested on two external datasets: the MIT-BIH AF database [17] and the MIMIC PERform AF dataset [55].

B. Experimentation Setup

All models were implemented in PyTorch [56] and trained on a high-performance setup with NVIDIA RTX 4080 GPUs (16 GB VRAM), a 12th Gen Intel Core i9-12900K CPU, and 128 GB RAM. The optimization was performed using the Adam optimizer, with a learning rate of 0.001 and a momentum of 0.9. To manage the learning rate, we implemented a “ReduceLROnPlateau” scheduler that reduces the learning rate by half when the validation loss plateaus. The scheduler has a patience of three epochs and a minimum learning rate of 1×10^{-6} .

TABLE I
OVERALL ARCHITECTURE DETAILS OF OUR PROPOSED MODEL (HYBRID COMPLEX AF-NET)

Block	C-Layer	Operations	P-Layer	(F, K, S)	Output Size
Input	P1	Complex Input	–	–	(224, 224, 3)
Block-A	A1	Conv2D + BN + ReLU	P1.real	(64, 7, 2)	(112, 112, 64)
	A2	Conv2D + BN + ReLU	P1.imag	(64, 7, 2)	(112, 112, 64)
Block-B	B1	Conv2D + BN + ReLU	A1	(64, 3, 1)	(112, 112, 64)
	B2	Conv2D + BN + ReLU	B1	(64, 3, 1)	(112, 112, 64)
	B3	Add (Residual)	B2, A1	–	(112, 112, 64)
Block-C	C1	Conv2D + BN + ReLU	B3	(64, 3, 1)	(112, 112, 64)
	C2	Conv2D + BN + ReLU	C1	(64, 3, 1)	(112, 112, 64)
	C3	Add (Residual)	B3, C2	–	(112, 112, 64)
Block-D	D1	Conv2D + BN + ReLU	A2	(64, 3, 1)	(112, 112, 64)
	D2	Conv2D + BN + ReLU	D1	(64, 3, 1)	(112, 112, 64)
	D3	Add (Residual)	D2, A2	–	(112, 112, 64)
Block-E	E1	Conv2D + BN + ReLU	D3	(64, 3, 1)	(112, 112, 64)
	E2	Conv2D + BN + ReLU	E1	(64, 3, 1)	(112, 112, 64)
	E3	Add (Residual)	E2, D3	–	(112, 112, 64)
Block-F	F1	ComplexConv2D + ComplexBN + ModReLU	C3 + i E3	(128, 3, 2)	(56, 56, 128)
	F2	ComplexConv2D + ComplexBN + ModReLU	F1	(128, 3, 1)	(56, 56, 128)
	F3	Identity Downsample (ComplexConv2D + ComplexBN)	C3 + i E3	(128, 1, 2)	(56, 56, 128)
	F4	Add (Residual)	F2, F3	–	(56, 56, 128)
Block-G	G1	ComplexConv2D + ComplexBN + ModReLU	F4	(128, 3, 1)	(56, 56, 128)
	G2	ComplexConv2D + ComplexBN + ModReLU	G1	(128, 3, 1)	(56, 56, 128)
	G3	Add (Residual)	F4, G2	–	(56, 56, 128)
Block-H	H1	Conv2D + BN + ReLU	G3.real	(256, 3, 2)	(28, 28, 256)
	H2	Conv2D + BN + ReLU	H1	(256, 3, 1)	(28, 28, 256)
	H3	Identity Downsample (Conv2D + BN)	G3.Real	(256, 1, 2)	(28, 28, 256)
	H4	Add (Residual)	H2, H3	–	(28, 28, 256)
Block-I	I1	Conv2D + BN + ReLU	H4	(256, 3, 1)	(28, 28, 256)
	I2	Conv2D + BN + ReLU	I1	(256, 3, 1)	(28, 28, 256)
	I3	Add (Residual)	I2, H4	–	(28, 28, 256)
Block-J	J1	Conv2D + BN + ReLU	G3.imag	(256, 3, 2)	(28, 28, 256)
	J2	Conv2D + BN + ReLU	J1	(256, 3, 1)	(28, 28, 256)
	J3	Identity Downsample (Conv2D + BN)	G3.imag	(256, 1, 2)	(28, 28, 256)
	J4	Add (Residual)	J2, J3	–	(28, 28, 256)
Block-K	K1	Conv2D + BN + ReLU	J4	(256, 3, 1)	(28, 28, 256)
	K2	Conv2D + BN + ReLU	K1	(256, 3, 1)	(28, 28, 256)
	K3	Add (Residual)	K2, J4	–	(28, 28, 256)
Block-L	L1	ComplexConv2D + ComplexBN + ModReLU	I3 + i K3	(512, 3, 2)	(14, 14, 512)
	L2	ComplexConv2D + ComplexBN + ModReLU	L1	(512, 3, 1)	(14, 14, 512)
	L3	Identity Downsample (ComplexConv2D + ComplexBN)	I3 + i K3	(512, 1, 2)	(14, 14, 512)
	L4	Add (Residual)	L2, L3	–	(14, 14, 512)
Block-M	M1	ComplexConv2D + ComplexBN + ModReLU	L4	(512, 3, 1)	(14, 14, 512)
	M2	ComplexConv2D + ComplexBN + ModReLU	M1	(512, 3, 1)	(14, 14, 512)
	M3	Add (Residual)	M2, L4	–	(14, 14, 512)
Output block	N1	Adaptive Avg Pool	M3	–	(1, 1, 512)
	N2	Magnitude	N1	–	(1, 1, 512)
	N3	Flatten	N2	–	512
	N3	Dense	N2	–	1

Note: F, K, and S denote the number of filters, kernel sizes, and strides used in each convolution layer, respectively. C- and P-layer denote the current and previous layers.

C. Results

To assess C-AFNet's performance on AF classification, we report accuracy, precision, recall, and F1-score, all computed from the confusion matrix shown in Fig. 5. The model converges rapidly, surpassing 90% accuracy within the first few epochs and maintaining stable performance. The loss curves indicate effective learning with minimal overfitting. On the validation set, C-AFNet achieves state-of-the-art 95.63% accuracy, 96.43% precision, 92.05% recall, and a 94.15% F1-score on the test set, demonstrating strong generalization.

D. Explainable Artificial Intelligence (AI) & Model Interpretability

To interpret the model's decisions, we employ Grad-CAM [57] and IGs [58]. Grad-CAM highlights important spatial regions in GAF images by leveraging gradients from the final convolutional layer, while IG attributes predictions by integrating gradients from a baseline to the input. To improve attribution stability, we apply the noise tunnel technique [59], which averages results across multiple noise-perturbed inputs. Fig. 6 illustrates the attribution maps for representative normal

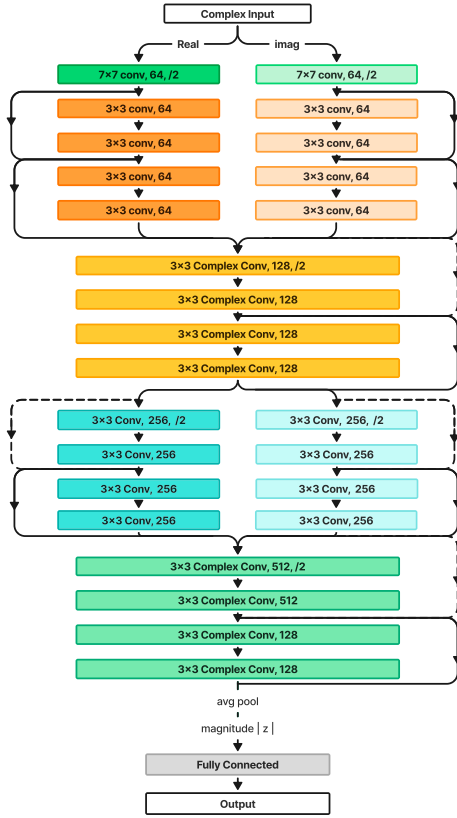


Fig. 4. Complete architecture of the proposed hybrid complex AF-Net.

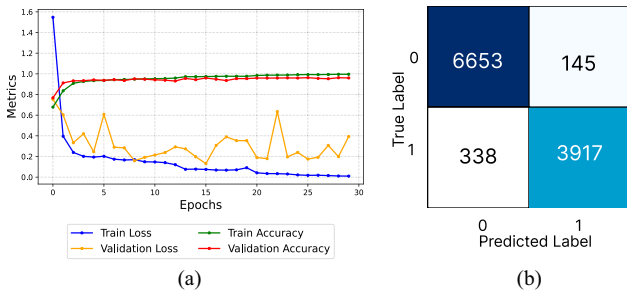


Fig. 5. C-AFNet performance. (a) Training/validation loss and accuracy over epochs. (b) Confusion matrix on the test set, where 0 belongs to normal and 1 belongs to AF.

and AF samples. For normal rhythm, the saliency maps show localized activations primarily within the GADF representation. Conversely, for AF cases, the model's attention shifts toward the GASF representation, highlighting irregular diagonal patterns. IGs with noise tunnel produce localized saliency maps for C-AFNet, while the ResNet-18 baseline demonstrates more diffuse attribution patterns across the input fields.

E. Computational Complexity Analysis

We assess the computational complexity of C-AFNet and baseline models by comparing total floating-point operations (FLOPs) and trainable parameters. FLOPs estimate inference

time by measuring addition, multiplication, and pooling operations. FLOPs are reported in giga-FLOPs (GFLOPs), where 1 GFLOP = 1 billion operations. Table II summarizes the FLOPs and parameter comparisons, highlighting C-AFNet's performance and computational cost tradeoffs.

F. Comparative Evaluation With Baseline Models

To benchmark our proposed complex-valued CNNs for AF classification, we designed a comprehensive evaluation setup encompassing both traditional and deep learning baselines. Traditional models including Naive Bayes [66], decision tree [67], K-nearest neighbors (KNN) [68], logistic regression [69], multi-layer perceptron (MLP) [71], long short-term memory (LSTM) [70], random forest [72], and SVM [73] were trained on handcrafted ECG features. These features were extracted after applying variational mode decomposition (VMD), which decomposed the signal into five intrinsic mode functions (IMFs), each yielding seven statistical features.

For deep learning baselines, we used GAF-transformed ECG images as input. Models such as ResNet-18 [38], ViT-B/16 [65], and MobileNetV3-Large (MBNetV3-L) [60] were trained end-to-end from scratch, while DenseNet-121 [63], EfficientNet-B0 [61], AlexNet [62], and VGG16 [64] were fine-tuned by updating only the classification layers. Models were trained on either GASF, GADF, or a combined six-channel GASF+GADF input [74]. As shown in Table III, our proposed C-AFNet outperforms all baselines, achieving 95.63% accuracy and 94.15% F1-score. To mathematically validate these gains, we conducted a statistical significance analysis using McNemar's tests and 95% bootstrap confidence intervals (CIs). C-AFNet outperformed the ResNet-18 (GADF) baseline with a McNemar significance of $p < 0.001$. Furthermore, C-AFNet achieved a Macro F1 CI of [0.9493, 0.9577], which is significantly higher than the [0.9320, 0.9413] recorded for ResNet-18. While the performance gap between C-AFNet and the ablated R-AFNet was narrower on the primary dataset ($p = 0.4661$), their respective 95% Accuracy CIs [0.9524, 0.9602] for C-AFNet and [0.9510, 0.9587] for R-AFNet demonstrate that the hybrid complex design maintains high stability even within the training distribution.

G. Generalization to External Datasets

To evaluate generalizability, C-AFNet was validated on two geographically and technically diverse datasets: the MIMIC PERform AF dataset [55] (1442 balanced samples) and the MIT-BIH AF database [17] (71 115 samples). As shown in Table IV, the ResNet18 baseline achieved 76.65% accuracy on the MIT-BIH set, while its AF recall was 8.99%, failing to detect the minority clinical target. In contrast, C-AFNet maintains an AF recall ($> 75\%$) across both domains. McNemar's tests ($p < 0.001$) and nonoverlapping 95% bootstrap CIs for Macro F1 (C-AFNet: [72.59%, 76.76%] versus ResNet-18: [39.71%, 43.80%] on MIMIC) were recorded.

H. Evaluation of Custom Metrics

In our experiments, we empirically set $\alpha = 0.78$, $\beta = 0.16$, $\gamma = 0.05$, $\lambda = 0.5$, and $\delta = 5$. We validate the proposed FS metric through both synthetic and empirical evaluations. Synthetic

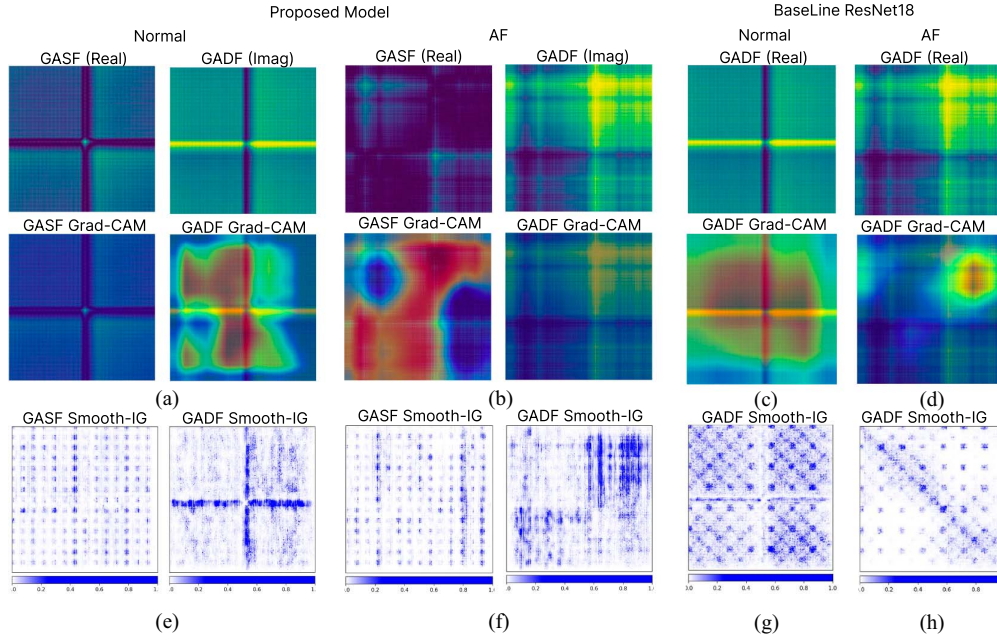


Fig. 6. Grad-CAM and IGs visualizations produced by ResNet18 and the proposed model for normal and AF cases. (a) Normal Grad-CAM. (b) AF Grad-CAM. (c) Normal Grad-CAM. (d) AF Grad-CAM. (e) Normal IG. (f) AF IGs (IG). (g) Normal IG. (h) AF IG. Note: AF = atrial fibrillation, IG = integrated gradients.

TABLE II
COMPARISON OF COMPUTATIONAL COMPLEXITY AMONG MODELS

Model	FLOPs (GFLOPs)	Parameters (M)
Real CNN and Transformer Models		
MobileNetV3-Large [60]	0.22	4.2
EfficientNet-B0 [61]	0.40	4.0
AlexNet [62]	0.71	57.0
ResNet18 [38]	1.82	11.2
DenseNet-121 [63]	2.86	7.0
VGG16 [64]	15.45	100.0
ViT-B/16 [65]	16.86	85.8
Our Proposed Complex Model and its Variants		
C-ResNet18	7.26	22.4
Inverted C-AFNet	5.62	22.4
C-AFNet (Maxpool)	5.28	22.4
Grouped C-AFNet	21.05	22.9
R-AFNet	13.83	22.4
Proposed (C-AFNet)	20.41	22.4

validation curves simulate diverse training behaviors such as smooth convergence, overfitting, and late learning. As shown in Table V and Fig. 7, the metric assigns higher scores to stable, generalizable patterns (e.g., ideal and optimal) and lower scores to erratic ones (e.g., mid training collapse and unstable learning). Fast learning is rewarded for early convergence via LS, while noisy or delayed patterns are penalized despite achieving good final accuracy. These results confirm the metric's alignment with intuitive training dynamics.

8) *Qualitative Analysis:* We performed a sensitivity analysis to select optimal weights α , β , and γ for the FS. Table VI shows

that $\alpha = 0.78$, $\beta = 0.16$, and $\gamma = 0.05$ yield the strongest correlation with test accuracy (Spearman: 0.8571, Pearson: 0.8125). Other settings, such as $\alpha = 0.6$, $\gamma = 0.4$, show reduced correlation, likely due to overemphasizing convergence speed over stability. These results confirm that our metric reliably reflects training quality and supports robust model selection.

I. Comparison of All Models Using Custom Metrics

Table VII summarizes the performance of all models based on the proposed metric. C-AFNet achieves the highest FS score (0.8928), attributed to its strong stability (SM = 0.9413) and generalization (OP = 0.9869), despite a moderate convergence speed (LS = 0.0144). Grouped C-AFNet closely follows with FS of 0.8920, benefiting from a perfect OP of 1.0000, though its lower LS (0.0029) indicates slower training dynamics. Among the baseline models, ResNet18 (GADF) performs best, with an FS of 0.8769, supported by a high SM (0.9242) and OP (0.9671). In contrast, C-ResNet18 and Inverted C-AFNet yield lower FS (0.8469 and 0.8468, respectively), primarily due to reduced generalization and slower convergence. These results emphasize the effectiveness of C-AFNet in balancing among the three evaluation criteria, underscoring its potential for robust and generalizable AF detection using GAF representations.

J. Ablation Study

To evaluate the specific design choices of our architecture, we conducted an ablation study comparing C-AFNet with several structural variants: Complex ResNet-18 (C-ResNet18), Inverted C-AFNet, Grouped C-AFNet, C-AFNet (MaxPool), and R-AFNet. C-ResNet18 is a fully complex-valued version of ResNet-18, applying complex operations throughout. Inverted

TABLE III
PERFORMANCE COMPARISON OF DIFFERENT MODELS

Model	Val	Test	Prec	Rec	F1
Traditional Models					
Naive Bayes [66]	39.58	42.36	59.55	42.36	37.42
Decision Tree [67]	67.36	62.50	63.22	62.50	62.81
KNN [68]	80.56	68.75	68.08	68.75	68.33
Logistic Regression [69]	77.08	73.61	72.74	73.61	72.13
LSTM [70]	82.64	73.61	73.01	73.61	73.18
MLP [71]	82.64	73.61	73.18	73.61	73.34
Random Forest [72]	81.94	75.69	75.24	75.69	74.21
SVM [73]	81.94	76.39	75.97	76.39	75.07
Real CNN and Transformer Models					
AlexNet (SF) [62]	78.07	72.99	71.08	50.31	58.92
AlexNet (DF) [62]	80.57	76.5	75.75	57.27	65.23
VGG16 (SF) [64]	80.26	72.41	70.92	48.03	57.27
VGG16 (DF) [64]	82.14	78.49	77.37	62.37	69.06
EfficientNet-B0 (SF) [61]	80.2	75.17	71.25	59.52	64.86
EfficientNet-B0 (DF) [61]	82.05	76.30	76.64	55.29	64.24
DenseNet-121 (SF) [63]	84.24	75.06	76.05	51.42	61.35
DenseNet-121 (DF) [63]	83.17	77.81	79.10	57.57	66.64
ViT-B/16 (SF) [65]	93.29	92.44	91.11	88.01	89.50
ViT-B/16 (DF) [65]	90.97	90.46	89.16	81.90	85.30
ResNet18 (SF) [38]	92.04	89.42	91.45	80.01	85.34
ResNet18 (SF+DF) [38]	93.95	90.47	93.50	80.86	86.72
ResNet18 (DF) [38]	95.23	94.11	95.75	88.62	92.05
MBNetV3-L (SF) [60]	94.76	92.92	94.15	87.02	90.44
MBNetV3-L (DF) [60]	95.48	94.69	95.62	90.64	92.91
Our Proposed Complex Model and its Variants					
C-ResNet18	93.41	92.02	92.68	86.06	89.25
Inverted C-AFNet	94.26	92.74	93.54	88.62	92.05
Grouped C-AFNet	95.40	94.73	96.15	89.92	92.93
C-AFNet (Maxpool)	95.51	94.27	95.34	89.49	92.32
R-AFNet	95.97	95.49	96.03	92.10	94.03
Proposed (C-AFNet)	96.28	95.63	96.43	92.05	94.19

Note: Val and Test denote validation and test accuracy, respectively; Prec, Rec, F1, DF, and SF refer to precision, recall, F1-score, GADF, and GASF. Bold entries indicate the highest/best performance metrics in the respective evaluations.

C-AFNet flips C-AFNet's design by using complex blocks in stages 1 and 3, and real blocks in stages 2 and 4. Grouped C-AFNet uses complex layers in stages 1 and 2, followed by real-valued layers in the latter half, and includes an attention gate aggregating features from final real-valued blocks. The MaxPool variant introduces early downsampling before feature extraction to study its impact. R-AFNet serves as the "ablated real" version of our backbone, utilizing the same staging as C-AFNet but replacing all complex operations with standard real-valued applied separately to the real and imaginary components. As shown in Tables III and IV, C-AFNet achieves the highest performance with 95.63% accuracy and a 94.15% F1-score. While R-AFNet shows comparable accuracy on the primary dataset (95.49%), C-AFNet achieves an AF F1-score of 76.95% compared to R-AFNet's 68.49%. These results are supported by McNemar's tests ($p < 0.001$) and nonoverlapping 95% bootstrap CIs for Macro F1 (C-AFNet: [72.59%, 76.76%] versus R-AFNet: [65.82%, 70.12%]). Furthermore, C-AFNet outperformed C-ResNet18 (92.02%) and Inverted C-AFNet (92.74%).

TABLE IV
CLASSWISE PERFORMANCE COMPARISON ACROSS
PRIMARY AND SECONDARY DATASETS

Model	Class	Prec	Rec	F1
Primary AF Dataset [26]				
ResNet18 (GADF)	AF	95.76	88.63	92.05
R-AFNet	AF	96.03	92.10	94.03
Proposed (C-AFNet)	AF	96.43	92.06	94.19
ResNet18 (GADF)	N	93.20	97.54	95.32
R-AFNet	N	95.18	97.62	96.38
Proposed (C-AFNet)	N	95.17	97.87	96.50
MIMIC PERform AF Dataset [55]				
ResNet18 (GADF)	AF	66.98	8.65	15.32
R-AFNet	AF	65.52	71.74	68.49
Proposed (C-AFNet)	AF	69.23	86.60	76.95
ResNet18 (GADF)	N	92.86	96.00	68.18
R-AFNet	N	70.93	64.61	67.62
Proposed (C-AFNet)	N	83.58	63.93	72.45
MIT-BIH AF Database [17]				
ResNet18 (GADF)	AF	24.26	8.99	13.12
R-AFNet	AF	36.19	73.84	48.57
Proposed (C-AFNet)	AF	39.24	75.42	51.62
ResNet18 (GADF)	N	80.76	93.15	86.51
R-AFNet	N	91.45	68.25	78.17
Proposed (C-AFNet)	N	92.27	71.52	80.58

Note: Bold entries indicate the highest/best performance metrics in the respective evaluations.

TABLE V
EVALUATING SYNTHETIC PATTERNS WITH CUSTOM METRICS

Pattern	SM	OP	LS	FS
Ideal case	0.9000	1.0000	0.0810	0.8661
Optimal Curve	0.8506	1.0000	0.0486	0.8259
Fast learning	0.8191	1.0000	0.0540	0.8016
Linear growth	0.6929	1.0000	0.0027	0.7006
Noisy Growth	0.6883	1.0000	0.0028	0.6970
Early saturation	0.6343	1.0000	0.0488	0.6572
Late surge	0.6104	1.0000	0.0027	0.6362
Linear Decline	0.6929	0.1353	0.0810	0.5662
Random walk	0.5964	0.5586	0.0586	0.5575
Worst case	0.5000	1.0000	0.0450	0.5522
Unstable Learning	0.5950	0.3679	0.0630	0.5261
Mid training collapse	0.5657	0.2466	0.0374	0.4826

K. Discussion

The observed performance gap confirms that unified complex representations preserve amplitude-phase coupling more effectively than real-valued fusion. Class-specific attention patterns observed via Grad-CAM and IGs align with cardiac physiology. The reliance on GADF for normal sinus rhythm reflects the regular phase patterns encoded by phase-difference embeddings, whereas the shift to GASF during AF episodes prioritizes the chaotic amplitude fluctuations characteristic of fibrillatory waves. Unlike the diffuse attributions of real-valued baselines, C-AFNet's sharp, localized features demonstrate that complex-valued layers identify specific morphological signatures essential for clinical trust in automated diagnosis.

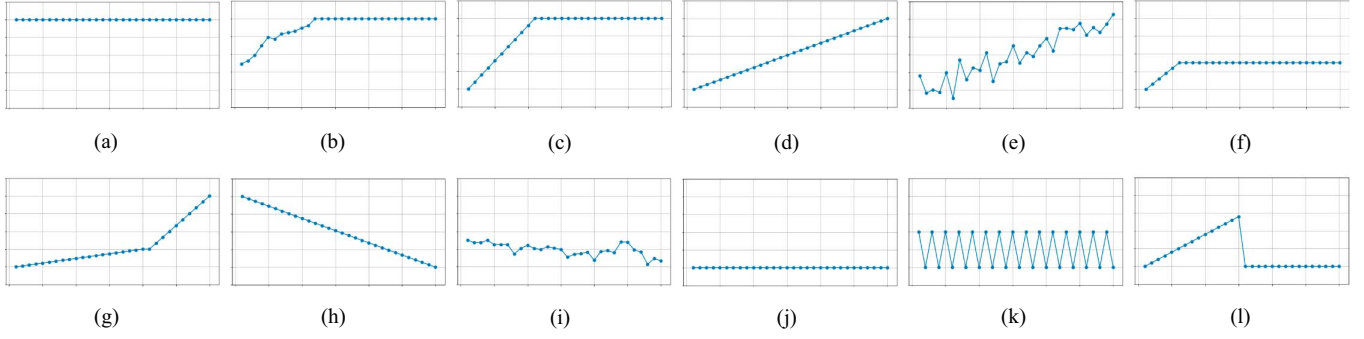


Fig. 7. Representative synthetic training accuracy curves (accuracy versus epochs) used to evaluate the metric in Table V. Each curve simulates a distinct learning behavior on which the metric is evaluated. The metric rewards smooth, monotonic growth and penalizes noisy or unstable patterns. (a) Ideal case. (b) Optimal curve. (c) Fast learning. (d) Linear growth. (e) Noisy growth. (f) Early saturation. (g) Late surge. (h) Linear decline. (i) Random walk. (j) Worst case. (k) Unstable learning. (l) Mid training collapse.

TABLE VI
CORRELATION OF FS AND ACCURACY UNDER VARYING PARAMETERS

α	β	γ	Spearman Correlation	Pearson Correlation
0.5	0.3	0.2	0.6786	0.7177
0.6	0.2	0.2	0.7143	0.7418
0.6	0.2	0.4	0.7143	0.6425
0.76	0.13	0.1	0.7143	0.7686
0.78	0.16	0.05	0.8571	0.8125
0.8	0.1	0.1	0.7500	0.8001

Note: Bold entries indicate the highest/best performance metrics in the respective evaluations.

TABLE VII
COMPARISON OF ALL MODELS USING CUSTOM METRICS

Model	SM	OP	LS	FS
ResNet18 (GASF)	0.8836	0.9738	0.0304	0.8465
Inverted C-AFNet (Ours)	0.8843	0.9799	0.0057	0.8468
C-ResNet18 (Ours)	0.8900	0.9525	0.0056	0.8469
ResNet18 (G + G)	0.9033	0.9757	0.0366	0.8625
ResNet18 (GADF)	0.9242	0.9671	0.0257	0.8769
Grouped C-AFNet (Ours)	0.9383	1.0000	0.0029	0.8920
Proposed (C-AFNet)	0.9413	0.9869	0.0144	0.8928

Note: Higher forecast stability indicates better generalization potential. Bold entries indicate the highest/best performance metrics in the respective evaluations.

Cross-dataset validation reveals that real-valued models often suffer from an accuracy paradox on imbalanced data. While the ResNet-18 baseline achieves high overall accuracy on the MIT-BIH dataset, its failure to detect the minority clinical target (8.99% recall) contrasts with C-AFNet's stable recall ($> 75\%$). This suggests that the hybrid complex design captures universal morphological features invariant to sensor-specific noise. Furthermore, the significant performance margin between C-AFNet and the ablated R-AFNet on external datasets ($p < 0.001$) confirms that independent real-valued paths cannot approximate the interdependent phase interactions captured by complex algebra.

The ablation study justifies the hybrid staging of real and complex layers. The failure of fully complex variants to match

TABLE VIII
PERFORMANCE COMPARISON OF PROPOSED MODEL WITH DIFFERENT MODELS

Authors	Acc.	Prec.	Rec.	F1
Kennedy et al. [14]	—	92.10	92.80	—
Rao et al. [15]	99.10	—	—	—
Hannun et al. [16]	—	—	—	83.7
O. Faust et al. [18]	98.51	—	—	—
S Mausavi et al. [19]	99.53	—	—	—
J Xie et al. [20]	88.65	—	—	—
Sun et al. [21]	91.32	—	—	89.41
D Kraft et al. [22]	—	99.0	98.6	98.6
Li et al. [23]	99.20	—	—	—
Hu et al. [24]	98.15	—	—	98.5
Gündüz & Tal [25]	87.65	—	—	—
Xie et al. [20]	—	—	—	88.65
Rahman et al. [27]	—	—	—	87.00
Aldughayfiq et al. [28]	95.0	88.00	85.00	84.00
Ben-Moshe et al. [29]	—	—	96.00	—
Ben-Moshe et al. [30]	—	—	—	96.00
Proposed (C-AFNet)	95.63	96.43	92.05	94.19

Note: Bold entries indicate the highest/best performance metrics in the respective evaluations.

C-AFNet's performance suggests that utilizing real-valued layers for initial spatial extraction followed by complex layers for phase refinement provides an optimal balance for hierarchical signal processing. Crucially, while C-AFNet and R-AFNet maintain an identical parameter count of 22.4 M, the operations involving complex values require a greater computational expenditure (20.41 GFLOPs compared to 13.83 GFLOPs). This gap suggests that performance improvements stem from enhanced representational capacity and mathematical profundity rather than architectural expansion. This arrangement guarantees the stability required for generalization across many therapeutic settings.

V. ADVANTAGES AND DRAWBACKS

The primary advantages of C-AFNet lie in its phase-aware modeling and clinical robustness, which provide a biologically grounded representation of cardiac depolarization. By natively capturing the interdependent relationship between signal magnitude and phase, the model avoids the recall collapse common in real-valued architectures when faced with technically diverse

hardware or geographically distant datasets. Furthermore, the model's high FS score (0.8928) and physiologically aligned saliency maps which focus on GADF for regular rhythms and GASF for arrhythmias ensure a reliable balance between training convergence and diagnostic interpretability that is essential for real-world applications.

Conversely, the model's drawbacks primarily involve computational complexity and preprocessing overhead associated with the 2-D GAF transformation and complex-valued operations. As evidenced by the 20.41 GFLOPs required, the architecture is more resource-intensive than lightweight 1-D CNNs, which may introduce inference latency and limit deployment on ultra-low-power edge devices. Additionally, the training pipeline is more intricate than standard real-valued frameworks, necessitating specialized weight initialization and dual-stream normalization to maintain gradient stability, which increases the overall implementation difficulty.

VI. CONCLUSION AND FUTURE WORKS

In this work, we proposed C-AFNet, a hybrid complex-real CNN for AF detection using GAF-encoded ECG signals. By integrating complex and real-valued layers in an alternate staged design, C-AFNet captures phase-aware temporal features and achieves state-of-the-art 95.63% accuracy. It outperforms other models like ResNet18, ViT, and MobileNetV3-Large in accuracy, F1-score, and training efficiency.

Beyond performance gains, this work contributes methodologically through improved training stability analysis and interpretable model behavior. More broadly, this study advances complex-valued deep learning for biomedical signal processing, demonstrating its effectiveness for ECG analysis and its potential applicability to other physiological signals such as EEG, EMG, and PPG. Future work includes optimizing C-AFNet for edge deployment, improving training stability through alternative optimization strategies, and extending the framework to multiclass arrhythmia classification. Additional directions involve large-scale clinical validation, multimodal integration, and self-supervised pre-training on unlabeled ECG data. Overall, C-AFNet represents a step toward more physiologically grounded, interpretable, and efficient deep learning models for cardiac arrhythmia detection.

REFERENCES

- [1] S. S. Chugh et al., "Worldwide epidemiology of atrial fibrillation: A global burden of disease 2010 study," *Circulation*, vol. 129, pp. 837–847, 2014.
- [2] D. K. Behera, D. B. Rahut, and S. Mishra, "Analyzing stroke burden and risk factors in India using data from the global burden of disease study," *Sci. Rep.*, vol. 14, no. 1, 2024, Art. no. 22640.
- [3] J.-H. Park et al., "Burden of stroke in the United States of America, 1990–2021: A systematic analysis for the us burden of disease study 2021," *Front. Neurol.*, vol. 16, 2025, Art. no. 1609508.
- [4] Y.-J. Wang et al., "China Stroke Statistics: Update on the 2019 report from National Neurological and Stroke Research Centers and Associations," *Stroke Vasc. Neurol.*, vol. 7, no. 5, pp. 415–450, 2022.
- [5] A. Callanan et al., "Atrial fibrillation (AF) pilot screening programme in primary care in Ireland: An implementation study protocol," *BMJ Open*, vol. 12, no. 2, 2022, Art. no. e054324.
- [6] G. Hindricks et al., "2020 ESC Guidelines for the diagnosis and management of atrial fibrillation, in collaboration with EACTS and with contribution from EHRA," *Eur. Heart J.*, vol. 42, pp. 373–498, 2021.
- [7] T. Chiheb et al., "Deep complex networks," in *Proc. Int. Conf. Learn. Represent.*, vol. 3, 2017.
- [8] J. Bassey, L. Qian, and X. Li, "A survey of complex-valued neural networks," 2021, *arXiv:2101.12249*.
- [9] N. Guberman, "On complex valued convolutional neural networks," 2016, *arXiv:1602.09036*.
- [10] E. Cole et al., "Analysis of deep complex-valued convolutional neural networks for MRI reconstruction and phase-focused applications," *Magn. Reson. Med.*, vol. 86, no. 2, pp. 1093–1109, 2021.
- [11] Z. Wang and T. Oates, "Encoding time series as images for visual inspection and classification using tiled convolutional neural networks," in *Proc. 29th AAAI Conf. Artif. Intell.*, vol. 1, 2015, pp. 91–96.
- [12] S. T. Aboyeji et al., "Feature fusion of Gramian angular field deep learning EEG-based epileptic seizure classification," in *Proc. 17th Int. Conv. Rehabil. Eng. Assistive Technol. (i-CREATE)*, Piscataway, NJ, USA: IEEE, 2024, pp. 1–6.
- [13] J. Qin and F. Liu, "GAF-FusionNet: Multimodal ECG analysis via Gramian angular fields and split attention," in *Proc. Int. Conf. Neural Inf. Process.*, 2024, pp. 299–312.
- [14] A. Kennedy et al., "Automated detection of atrial fibrillation using R-R intervals and multivariate-based classification," *J. Electrocardiol.*, vol. 49, no. 6, pp. 871–876, 2016.
- [15] S. K. S. Rao and R. J. Martis, "RR Interval-based atrial fibrillation detection using traditional and ensemble machine learning algorithms," *J. Med. Signals Sens.*, vol. 13, pp. 224–232, 2023.
- [16] A. Y. Hannun et al., "Cardiologist-level arrhythmia detection and classification in ambulatory electrocardiograms using a deep neural network," *Nat. Med.*, vol. 25, no. 1, pp. 65–69, 2019.
- [17] A. L. Goldberger et al., "PhysioBank, PhysioToolkit, and PhysioNet: Components of a new research resource for complex physiologic signals," *Circulation*, vol. 101, no. 23, pp. e215–e220, 2000.
- [18] O. Faust et al., "Automated detection of atrial fibrillation using long short-term memory network with RR interval signals," *Comput. Biol. Med.*, vol. 102, pp. 327–335, 2018.
- [19] S. Mousavi and F. Afghah, "Inter-and intra-patient ECG heartbeat classification for arrhythmia detection," in *Proc. IEEE Int. Conf. Acoust., Speech Signal Process.*, 2019, pp. 1308–1312.
- [20] J. Xie, S. Stavrakis, and B. Yao, "Automated identification of atrial fibrillation from single-lead ECGs using multi-branching ResNet," *Front. Physiol.*, vol. 15, p. 1362185, 2024.
- [21] L.-C. Sun et al., "Deep learning for the classification of atrial fibrillation using wavelet transform-based visual images," *BMC Med. Inform. Decis. Making*, vol. 22, no. S5, p. 349, 2025.
- [22] D. Kraft and P. Rumm, "Atrial fibrillation and atrial flutter detection using deep learning," *Sensors*, vol. 25, no. 13, 2025, Art. no. 4109.
- [23] X. Li et al., "SEResUTer: A deep learning approach for accurate ECG signal delineation and atrial fibrillation detection," *Physiol. Meas.*, vol. 44, p. 125005, 2023.
- [24] Y. Hu et al., "Detection of paroxysmal atrial fibrillation from dynamic ECG recordings based on a deep learning model," SSRN 4098696, 2023.
- [25] A. F. Gündüz and M. F. Talu, "Atrial fibrillation classification and detection from ECG recordings," *Biomed. Signal Process. Control*, vol. 82, p. 104531, 2023.
- [26] X. Wang et al., "Paroxysmal atrial fibrillation events detection from dynamic ECG recordings: The 4th China Physiological Signal Challenge 2021," in *Proc. PhysioNet*, 2021, pp. 1–83.
- [27] M. A. Rahman et al., "A deep learning scheme for detecting atrial fibrillation based on fusion of raw and discrete wavelet transformed ECG features," in *Proc. 44th Annu. Int. Conf. IEEE Eng. Med. Biol. Soc. (EMBC)*, 2022, pp. 1024–1027.
- [28] B. Aldughayfiq et al., "A deep learning approach for atrial fibrillation classification using multi-feature time series data from ECG and PPG," *Diagnostics*, vol. 13, 2023, Art. no. 2442.
- [29] N. Ben-Moshe et al., "RawECGnet: Deep learning generalization for atrial fibrillation detection from the raw ECG," *IEEE J. Biomed. Health Inform.*, vol. 28, no. 9, pp. 5180–5188, Sep. 2024.
- [30] N. Ben-Moshe, S. Biton, and J. A. Behar, "Arnet-ECG: Deep learning for the detection of atrial fibrillation from the raw electrocardiogram," in *Proc. Comput. Cardiol. (CinC)*, vol. 498, 2022, pp. 1–4.
- [31] S. Noorzadeh et al., "Modeling quasi-periodic signals by a non-parametric model: Application on fetal ECG extraction," in *Proc. 36th Annu. Int. Conf. IEEE Eng. Med. Biol. Society (EMBC)*, 2014, pp. 1889–1892.
- [32] S.-H. Noh, "Analysis of gradient vanishing of RNNs and performance comparison," *Information*, vol. 12, p. 442, 2021.

- [33] Z. Wang and T. Oates, "Imaging time-series to improve classification and imputation," 2015, *arXiv:1506.00327*.
- [34] Z. Ahmad et al., "ECG heart-beat classification using multimodal image fusion," in *Proc. Int. Conf. Acoust., Speech Signal Process. (ICASSP)*, 2021, pp. 1330–1334.
- [35] Y. Elmir, Y. Himeur, and A. Amira, "ECG classification using deep CNN and Gramian angular field," in *Proc. IEEE 9th Int. Conf. Big Data Comput. Service Appl. (BigDataService)*, 2023, pp. 137–141.
- [36] G.-W. Yoon and S. Joo, "Enhanced electrocardiogram classification using Gramian angular field transformation with multi-lead analysis and segmentation techniques," *MethodsX*, vol. 14, 2025, Art. no. 103297.
- [37] H. Xu et al., "Human activity recognition based on Gramian angular field and deep convolutional neural network," *IEEE Access*, vol. 8, pp. 199393–199405, 2020.
- [38] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, 2016, pp. 770–778.
- [39] S. Ioffe and C. Szegedy, "Batch normalization: Accelerating deep network training by reducing internal covariate shift," in *Proc. Int. Conf. Mach. Learn.*, 2015, pp. 448–456.
- [40] V. Nair and G. E. Hinton, "Rectified linear units improve restricted Boltzmann machines," in *Proc. 27th Int. Conf. Mach. Learn. (ICML-10)*, 2010, pp. 807–814.
- [41] K. He et al., "Delving deep into rectifiers: Surpassing human-level performance on ImageNet classification," in *Proc. IEEE Int. Conf. Comput. Vis.*, 2015, pp. 1026–1034.
- [42] S. Santurkar et al., "How does batch normalization help optimization?" in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 31, 2018.
- [43] J. S. Dramsch, M. L  thje, and A. N. Christensen, "Complex-valued neural networks for machine learning on non-stationary physical data," *Comput. Geosci.*, vol. 146, 2021, Art. no. 104643.
- [44] M. Arjovsky, A. Shah, and Y. Bengio, "Unitary evolution recurrent neural networks," in *Proc. Int. Conf. Mach. Learn.*, 2016, pp. 1120–1128.
- [45] J. A. Barrachina et al., "Theory and implementation of complex-valued neural networks," 2023, *arXiv:2302.08286*.
- [46] A. Paszke et al., "PyTorch: An imperative style, high-performance deep learning library," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 32, 2019.
- [47] I. Bello et al., "Revisiting ResNets: Improved training and scaling strategies," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 34, 2021, pp. 22614–22627.
- [48] A. Zafar et al., "A comparison of pooling methods for convolutional neural networks," *Appl. Sci.*, vol. 12, no. 17, 2022, Art. no. 8643.
- [49] O. Bousquet and A. Elisseeff, "Stability and generalization," *J. Mach. Learn. Res.*, vol. 2, no. Mar, pp. 499–526, 2002.
- [50] M. Hardt, B. Recht, and Y. Singer, "Train faster, generalize better: Stability of stochastic gradient descent," in *Proc. Int. Conf. Mach. Learn.*, 2016, pp. 1225–1234.
- [51] G. C. Cawley and N. L. Talbot, "On over-fitting in model selection and subsequent selection bias in performance evaluation," *J. Mach. Learn. Res.*, vol. 11, pp. 2079–2107, 2010.
- [52] I. Goodfellow, Y. Bengio, and A. Courville, "Regularization for deep learning," *Deep Learn.*, vol. 1, pp. 224–270, 2016.
- [53] L. N. Smith and N. Topin, "Super-convergence: Very fast training of neural networks using large learning rates," 2017, *arXiv:1708.07120*.
- [54] D. Mishkin and J. Matas, "All you need is a good in IT," 2015, *arXiv:1511.06422*.
- [55] P. H. Charlton et al., "Detecting beats in the photoplethysmogram: Benchmarking open-source algorithms," *Physiol. Meas.*, vol. 43, no. 8, 2022, Art. no. 085007.
- [56] G. Van Rossum and F. L. Drake, *Python 3 Reference Manual*. Scotts Valley, CA, USA: CreateSpace, 2009.
- [57] R. R. Selvaraju et al., "Grad-CAM: Visual explanations from deep networks via gradient-based localization," *Int. J. Comput. Vis.*, vol. 128, pp. 336–359, 2019.
- [58] M. Sundararajan, A. Taly, and Q. Yan, "Axiomatic attribution for deep networks," in *Proc. Int. Conf. Mach. Learn.*, 2017, pp. 3319–3328.
- [59] J. Adebayo, J. Gilmer, M. Muelly, I. Goodfellow, M. Hardt, and B. Kim, "Sanity checks for saliency maps," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 31, 2018.
- [60] A. Howard et al., "Searching for MobileNetV3," in *Proc. IEEE/CVF Int. Conf. Comput. Vis.*, 2019, pp. 1314–1324.
- [61] M. Tan and Q. Le, "EfficientNet: Rethinking model scaling for convolutional neural networks," in *Proc. Int. Conf. Mach. Learn.*, 2019, pp. 6105–6114.
- [62] M. Z. Alom et al., "The history began from AlexNet: A comprehensive survey on DL approaches," 2018, *arXiv:1803.01164*.
- [63] G. Huang et al., "Densely connected convolutional networks," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, 2017, pp. 4700–4708.
- [64] K. Simonyan and A. Zisserman, "Very deep convolutional networks for large-scale image recognition," 2014, *arXiv:1409.1556*.
- [65] A. Dosovitskiy et al., "An image is worth 16x16 words: Transformers for image recognition at scale," 2021, *arXiv:2010.11929*.
- [66] E. Miranda et al., "Detection of cardiovascular disease risk's level for adults using naive bayes classifier," *Healthcare Inform. Res.*, vol. 22, no. 3, pp. 196–205, 2016.
- [67] B. Charbuty and A. Abdulazeez, "Classification based on decision tree algorithm for machine learning," *J. Appl. Sci. Technol. Trends*, vol. 2, no. 1, pp. 20–28, 2021.
- [68] M. Abubaker and W. Ashour, "Efficient data clustering algorithms: Improvements over Kmeans," *Int. J. Intell. Syst. Appl.*, vol. 5, no. 3, 2013, Art. no. 37.
- [69] D. W. Hosmer, Jr., S. Lemeshow, and R. X. Sturdivant, *Applied Logistic Regression*. Hoboken, NJ, USA: Wiley, 2013.
- [70] S. Hochreiter and J. Schmidhuber, "Long short-term memory," *Neural Comput.*, vol. 9, no. 8, pp. 1735–1780, 1997.
- [71] M. W. Gardner and S. R. Dorling, "Artificial neural networks a review of applications in the atmospheric sciences," *Atmos. Environ.*, vol. 32, no. 14–15, pp. 2627–2636, 1998.
- [72] L. Breiman, "Random forests," *Mach. Learn.*, vol. 45, pp. 5–32, 2001.
- [73] M. A. Hearst, S. T. Dumais, E. Osuna, J. Platt, and B. Scholkopf, "Support vector machines," *IEEE Intell. Syst. Their Appl.*, vol. 13, no. 4, pp. 18–28, Jul./Aug. 1998.
- [74] S. Paheding et al., "GAF-NAU: Gramian angular field encoded neighborhood attention u-net for pixel-wise hyperspectral image classification," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, 2022, pp. 409–417.